

Viper Applied Science

Viper Blast

Theory Manual

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1 Introduction

VIPER BLAST (herein VIPER) is a computational fluid dynamics (CFD) based tool for simulating the airblast environment resulting from detonation of high explosives and other sources, such as vapours and high pressure gas. The primary purpose of VIPER is to predict blast loads on structures of non-trivial geometry.

Two important aspects of VIPER are the graphical user interface (GUI) and the hardware acceleration of the numerical scheme. The GUI is designed for ease-of-use with a short learning curve and is intended for use within a blast engineering consultancy environment where rapid project turnaround is required. Implementation of the numerical scheme has been targeted at NVIDIA GPUs (Graphic Processing Units) in order to further reduce analysis times and project costs.

This theory manual is intended for primarily internal use and for use by the enthusiastic customer who wants an in-depth understanding of VIPER. This will be a comprehensive and detailed guide of the physics and numerical implementation. For specifics on how to use VIPER, we refer the reader to the *User Manual*.

The basic functionality of VIPER is based largely upon the methodology described by Rose [2001] and the Air3D manual [Rose, 2006] and uses the AUSMDV numerical scheme as described by Wada and Liou [1994].

2 Continuum equations

To model air blasts, we solve the Euler equations. This includes the continuity equation,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{V}) = 0, \tag{2.1}$$

the momentum equations,

$$\frac{\partial (\rho u)}{\partial t} + \nabla \cdot (\rho u \boldsymbol{V}) + \frac{\partial p}{\partial x} = 0, \tag{2.2a}$$

$$\frac{\partial (\rho v)}{\partial t} + \nabla \cdot (\rho v \boldsymbol{V}) + \frac{\partial p}{\partial y} = 0, \tag{2.2b}$$

$$\frac{\partial (\rho w)}{\partial t} + \nabla \cdot (\rho w \boldsymbol{V}) + \frac{\partial p}{\partial z} = 0, \tag{2.2c}$$

and the total energy equation,

$$\frac{\partial}{\partial t}\left[\rho\left(e + \frac{V^2}{2}\right)\right] + \nabla \cdot \left[\rho\left(e + \frac{V^2}{2}\right) + p\right]\boldsymbol{V} = 0, \tag{2.3}$$

where ρ is density, $\boldsymbol{V}$ is the velocity vector with components u, v, and w, in the $x-$, $y-$, and $z-$directions, respectively, p is the pressure, and e is the specific internal energy. For simplicity, we define specific total energy to be

$$E = e + \frac{V^2}{2}. \tag{2.4}$$

Finally, we also require an equation of state (see Section 2.1) for closure.

For succinctness, these equations are generally condensed to a single equation. In one dimension,

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}}{\partial x} = \mathbf{0} \tag{2.5}$$

and in three dimensions,

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}_{\mathrm{x}}}{\partial x} + \frac{\partial \mathbf{F}_{\mathrm{y}}}{\partial y} + \frac{\partial \mathbf{F}_{\mathrm{z}}}{\partial z} = \mathbf{0}. \tag{2.6}$$

Here, $\mathbf{U}$ is the complete system of *conserved variables* and $\mathbf{F}$ is *flux vectors*. In three-dimensions, these vectors are

given by

$$\mathbf{U} = \begin{Bmatrix} \rho \\ \rho u \\ \rho v \\ \rho w \\ \rho\left(e + \frac{V^2}{2}\right) \end{Bmatrix}, \tag{2.7}$$

$$\mathbf{F}_\mathrm{x} = \begin{Bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ \rho uw \\ \rho u\left(e + \frac{V^2}{2}\right) + pu \end{Bmatrix}, \mathbf{F}_\mathrm{y} = \begin{Bmatrix} \rho v \\ \rho vu \\ \rho v^2 + p \\ \rho uw \\ \rho v\left(e + \frac{V^2}{2}\right) + pv \end{Bmatrix}, \mathbf{F}_\mathrm{z} = \begin{Bmatrix} \rho w \\ \rho wu \\ \rho wv \\ \rho w^2 + p \\ \rho w\left(e + \frac{V^2}{2}\right) + pw \end{Bmatrix}. \tag{2.8}$$

To decrease from three- to one-dimension, only the first, second, and fifth rows of $\mathbf{U}$ and $\mathbf{F}_\mathrm{x}$ are required.

2.1 Equations of state

Currently, VIPER has two equations of state, ideal gas and Jones-Wilkins-Lee (JWL). Each solver (see below) is hard-coded with one of the two equations of state for efficiency.

The ideal equation of state is given by

$$p = (\gamma - 1)\rho e \tag{2.9}$$

where $\gamma = 1.4$ is the (hard-coded) adiabatic index for air. The fixed γ means that the results become less accurate at high temperatures, but the results are still typically within reasonable uncertainty when comparing to experiments. In many models and many regimes, this is an acceptable approximation.

When ideal gas may not fully represent the scenario being modelled, the JWL [Jones and Miller, 1948, Wilkins et al., 1964, Lee et al., 1968] equation of state[1] can be used. This EOS describes the thermodynamics of both detonation products and unreacted high explosives, and retains accuracy at high temperatures. This roughly takes the form of

$$p = A\left(1 - \frac{\omega}{R_1 V}\right)e^{-R_1 V} + B\left(1 - \frac{\omega}{R_2 V}\right)e^{-R_2 V} + \frac{\omega E}{V}, \tag{2.10a}$$

$$p(\Delta) = Ae^{-R_1 V} + Be^{-R_2 V} + \frac{C}{V^{\omega+1}}, \tag{2.10b}$$

where A, B, C, E, R_1 and R_2, are parameters that depend on the type of explosive. These parameters are emperically determined by comparison to experiments, thus inherently include uncertainty from the experiments. VIPER includes default values for the included explosives, but they can be be modified by the user during initialisation. For further details and definition of the terms, see eqn 1.3 and associated text of of Lee et al. [1968].

3 Numerical methods

VIPER includes several user options, including dimensionality, boundaries, equations of state, all of which can be modified in the GUI prior to execution.

3.1 Scheme

VIPER evolves the fluid equations (2.6) using the AUSMDV scheme [Wada and Liou, 1994]. This is a combination of the AUSMD & AUSMV schemes, which themselves are based upon the Advection Upstream Splitting Method [AUSM; Liou and Steffen, 1993]. The sheme in VIPER is second order in both time and space, and is based upon that described in section 2 of Rose [2001]; it has since been improved for enhanced stability and performance. For example, VIPER includes corrections for the Carbuncle instability and includes an option for improved stability when evolving the blast into the far field.

[1] The JWL EOS is named and completely described in Lee et al. [1968], and builds upon Jones and Miller [1948] and Wilkins et al. [1964].

The general scheme effectively treats each dimension independently, and splits each dimension into two phases. In the first phase, the conserved variables at the centre of the cells are used to calculate the conserved values at the edge of the cells at the half-timestep. For each cell, this calculation is performed using a template of three cells, which comprises of the the given cell and cells on either side. This yields two values at each edge, since each cell uses a different template of three cells. Then in the second phase, edge values are first updated to produce a single edge value, and then these new edge values are used to calculate the conserved values in the centre of the cells at the full timestep.

3.2 Boundary conditions

In VIPER, the active domain is surrounded by a layer of two boundary (or ghost) cells. These ghost cells are required to properly calculate the values in the cells near the edge of the active domain, but are not evolved themselves. Rather, at the end of each step, these ghost cells receive values from the neighbouring active cells depending on the type of boundary requested. VIPER includes three possible boundary conditions, which can be set individually for each boundary at runtime.

3.2.1 Transmitting

In transmitting boundaries, the ghost cells take on the values of their neighbours. Here,

$$U_{1,j,k} = U_{2,j,k} \tag{3.1a}$$
$$U_{0,j,k} = U_{3,j,k} \tag{3.1b}$$
$$U_{\mathrm{ncx}+2,j,k} = U_{\mathrm{ncx}+1,j,k} \tag{3.1c}$$
$$U_{\mathrm{ncx}+3,j,k} = U_{\mathrm{ncx}+0,j,k} \tag{3.1d}$$

for all U, where the values on the left-hand side of the expressions are the ghost cells and the values on the right-hand side are in the active domain; the active domain in the x direction extends from 2 to `ncx+1`. Similar expressions are used in the remaining two dimensions.

3.2.2 Reflecting

Reflecting boundaries are the same as transmitting, except that the momentum in the direction of evolution is the negative of the value in the active region. For example, in 3D, the reflecting boundaries for the first layer of ghost cells on the left-hand side of the domain are

$$\rho_{1,j,k} = \rho_{2,j,k} \tag{3.2a}$$
$$\rho v^{\mathrm{x}}_{1,j,k} = -\rho v^{\mathrm{x}}_{2,j,k} \tag{3.2b}$$
$$\rho v^{\mathrm{y}}_{1,j,k} = \rho v^{\mathrm{y}}_{2,j,k} \tag{3.2c}$$
$$\rho v^{\mathrm{z}}_{1,j,k} = \rho v^{\mathrm{z}}_{2,j,k} \tag{3.2d}$$
$$\rho E_{1,j,k} = \rho E_{2,j,k} \tag{3.2e}$$

Similar expressions are required for the remaining three pairs of indices as shown in Eqn. 3.1, and similar expressions can be constructed for the right-hand side of the domain and the remaining directions.

3.2.3 Terminating

Terminating boundaries are designed to end the simulation once the flow has reached the desired boundary. Thus, the fluid is only ever evolved in the active region and is only ever affected by physics within the active regions. The conditions for terminating boundaries are below.

If a terminating boundary is set in 1D, then the simulation will terminate if $\Delta\rho = |\rho_{\mathrm{ncx}+1} - \rho_{\mathrm{ncx}}| > 0.001$, where $\rho_{\mathrm{nxc}+1}$ and ρ_{ncx} are the final two cells in the active domain. Note that the boundary at $i = 2$ must be reflecting in 1D.

If a terminating boundary is set in 2D, then the simulation will terminate if the momentum becomes non-zero in any active cell next to the boundary, i.e., if $|\rho v_{\mathrm{x}}|_{i,j} > 0.01$ or $|\rho v_{\mathrm{y}}|_{i,j} > 0.01$ for any $(i,j) \in \{(2,j), (\mathrm{ncx}+2, j), (i,2), (j, \mathrm{ncy}+2)\}$.

Terminating boundaries are not available in 3D.

3.3 Obstacles

Viper permits obstacles to be included within the fluid domain. Fluid cells are defined as fully inside or fully outside of the obstacle, so the obstacle boundary is necessarily coincident with the cell edge.

3.4 Dimensionality

Viper can be run in one-, two-, and three-dimensions. Each solver is hard-coded to be run in one of these dimensions.

One dimension assumes spherical symmetry. The left-hand boundary is defined to be reflecting and the right-hand boundary it a termination boundary. The bomb is necessarily placed at the origin in one-dimension.

Two dimensions assume cylindrical symmetry. The axis of symmetry is defined to be reflecting while the boundaries at the remaining three edges can be selected by the user.

Three dimensions includes no assumptions on symmetry. Currently, the user can select reflecting or transmitting for each of the eight edges.

3.4.1 Remaps

For efficiency, the simulation can be paused and 'remapped' onto a higher-order grid by making use of the symmetry of the lower-dimension simulation. For example, an air burst can be modelled in 1D until just before the blast impacts the ground; just prior to impact, the simulation can be paused, remapped into 2D and then continued. This remapping is explained in the User Manual, and the (manual) work flow is designed for improved efficiency and accuracy since lower-dimensional simulations can be run at higher resolution and/or faster. No information is lost in the process as long as the remapping occurs prior to any symmetry-breaking, and so long as the remap is into ambient atomsphere.

3.5 Viscosity

There is no explicit viscosity, therefore turbulence can not be modelled. Artificial viscosity is not required for the AUSMDV method, thus not included; this is a departure from other advection schemes.

3.6 Afterburn

Afterburn is discussed in Example 23 of the User Manual. From a numerical perspective, at all times and for every cell that contains burned and/or unburned material, energy is added to the total energy of the cell. The amount of energy is calculated upon an 'S-curve', which is defined by an afterburn duration and total amount of energy to be added. Therefore, this curve dictates how much energy is to be added at each timestep by determining the difference in energies between the next timestep and the current step. Note that by the nature of the curve, at very small times and for times larger than the parameterised burn duration, the amount of energy added is negligible since this difference approaches zero.

4 Conclusion

We refer the reader to the *User Manual* for Examples and Verification & Validation of Viper itself.

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A Papers that use Viper

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